

## VELIKO TARNOVO, Bulgaria

WMO: 155300

Lat: 43.083N

Lon: 25.650E

Elev: 218

StdP: 98.73

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.0      | -9.0 | -15.3                         | 1.0 | -11.5 | -12.2 | 1.4 | -8.2 | 13.8                  | 7.4  | 10.2 | 6.4  | 0.6                   | 270  | 0.424 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 35.0            | 21.1 | 33.0 | 20.7 | 31.3 | 20.3 | 23.0                | 30.9 | 22.0 | 29.9 | 21.2 | 29.0 | 2.3                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.5                            | 15.6 | 26.7 | 19.5 | 14.7 | 25.4 | 18.7 | 13.9 | 24.4 | 69.1          | 31.0 | 65.5 | 29.7 | 62.4 | 29.1 | 29.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.7           | 6.8  | 5.3 | -15.9                      | 38.0 | 3.8                | 2.7 | -18.7                                              | 39.9 | -21.0      | 41.5 | -23.1      | 43.0 | -25.9      | 45.0 |    |    |
| (5)               |      |     | -16.5                      | 24.9 | 3.8                | 2.0 | -19.2                                              | 26.3 | -21.4      | 27.4 | -23.5      | 28.6 | -26.2      | 30.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.5   | 0.9   | 2.9   | 7.2   | 12.3  | 17.5  | 21.2  | 23.5  | 23.3  | 18.5  | 12.9  | 7.3   | 1.9   |  |  |
|                                                                        | DBStd       | 9.06   | 5.28  | 5.61  | 4.75  | 3.83  | 3.39  | 3.13  | 2.84  | 2.94  | 3.61  | 4.36  | 4.75  | 5.26  |  |  |
|                                                                        | HDD10.0     | 1009   | 286   | 202   | 113   | 21    | 1     | 0     | 0     | 0     | 0     | 22    | 108   | 256   |  |  |
|                                                                        | HDD18.3     | 2628   | 542   | 432   | 344   | 183   | 57    | 10    | 1     | 2     | 42    | 175   | 331   | 510   |  |  |
|                                                                        | CDD10.0     | 1919   | 3     | 3     | 27    | 89    | 232   | 336   | 417   | 414   | 254   | 111   | 28    | 4     |  |  |
|                                                                        | CDD18.3     | 497    | 0     | 0     | 0     | 2     | 30    | 96    | 160   | 157   | 46    | 6     | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 5356   | 0     | 0     | 4     | 36    | 352   | 977   | 1731  | 1696  | 484   | 74    | 3     | 0     |  |  |
|                                                                        | CDH26.7     | 2186   | 0     | 0     | 0     | 6     | 79    | 363   | 784   | 778   | 159   | 15    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.5    | 1.4   | 1.6   | 2.0   | 1.9   | 1.5   | 1.5   | 1.4   | 1.3   | 1.4   | 1.2   | 1.4   | 1.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 650    | 43    | 42    | 49    | 54    | 78    | 68    | 67    | 47    | 63    | 46    | 41    | 56    |  |  |
|                                                                        | PrecMax     | 1125   | 102   | 89    | 152   | 111   | 167   | 138   | 202   | 140   | 221   | 114   | 96    | 97    |  |  |
|                                                                        | PrecMin     | 423    | 10    | 6     | 3     | 3     | 23    | 9     | 4     | 4     | 3     | 1     | 2     | 8     |  |  |
|                                                                        | PrecStd     | 163    | 21    | 24    | 33    | 28    | 43    | 40    | 53    | 35    | 52    | 29    | 25    | 24    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 16.5  | 19.2  | 23.5  | 27.4  | 31.1  | 34.6  | 38.3  | 37.5  | 33.3  | 28.7  | 23.2  | 18.1  |  |  |
|                                                                        |             | MCWB   | 10.6  | 11.0  | 13.9  | 16.2  | 19.4  | 21.0  | 21.1  | 20.9  | 19.4  | 18.6  | 15.6  | 12.2  |  |  |
|                                                                        | 2%          | DB     | 14.1  | 16.3  | 20.4  | 23.7  | 28.8  | 32.4  | 35.3  | 35.0  | 30.5  | 25.4  | 20.2  | 15.0  |  |  |
|                                                                        |             | MCWB   | 9.5   | 9.9   | 12.0  | 14.4  | 17.9  | 21.4  | 21.2  | 21.3  | 18.5  | 17.0  | 14.8  | 10.5  |  |  |
|                                                                        | 5%          | DB     | 11.7  | 14.1  | 18.2  | 21.6  | 27.0  | 30.5  | 32.9  | 32.9  | 28.4  | 22.8  | 17.8  | 12.3  |  |  |
|                                                                        |             | MCWB   | 7.8   | 8.9   | 10.7  | 13.4  | 17.5  | 20.7  | 20.8  | 20.6  | 18.2  | 16.2  | 13.2  | 9.1   |  |  |
|                                                                        | 10%         | DB     | 8.7   | 11.6  | 15.7  | 19.8  | 25.2  | 28.7  | 30.9  | 31.1  | 26.0  | 20.4  | 15.2  | 9.8   |  |  |
|                                                                        |             | MCWB   | 5.9   | 7.5   | 9.6   | 12.5  | 16.9  | 19.9  | 20.5  | 20.1  | 17.4  | 14.9  | 11.6  | 7.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 11.1  | 11.6  | 14.4  | 16.8  | 20.6  | 23.6  | 24.2  | 24.3  | 20.6  | 19.2  | 16.9  | 12.5  |  |  |
|                                                                        |             | MCDB   | 15.6  | 18.0  | 21.6  | 25.2  | 28.3  | 30.6  | 32.8  | 31.5  | 27.5  | 26.8  | 20.7  | 17.0  |  |  |
|                                                                        | 2%          | WB     | 9.7   | 10.3  | 12.4  | 15.1  | 19.3  | 22.6  | 22.7  | 23.0  | 19.6  | 17.8  | 14.8  | 10.7  |  |  |
|                                                                        |             | MCDB   | 13.7  | 15.6  | 19.3  | 21.6  | 26.6  | 30.1  | 31.0  | 30.8  | 28.0  | 24.2  | 19.7  | 14.2  |  |  |
|                                                                        | 5%          | WB     | 8.1   | 9.0   | 11.2  | 14.1  | 18.4  | 21.5  | 21.9  | 21.8  | 18.9  | 16.5  | 13.4  | 9.2   |  |  |
|                                                                        |             | MCDB   | 11.5  | 13.8  | 17.3  | 19.9  | 25.1  | 28.9  | 29.7  | 29.8  | 26.7  | 21.7  | 17.4  | 12.2  |  |  |
|                                                                        | 10%         | WB     | 6.0   | 7.6   | 9.9   | 13.2  | 17.4  | 20.5  | 21.1  | 20.9  | 18.1  | 15.3  | 11.8  | 7.4   |  |  |
|                                                                        |             | MCDB   | 8.6   | 11.5  | 15.2  | 18.5  | 23.6  | 27.3  | 28.8  | 29.3  | 24.8  | 19.6  | 14.9  | 9.5   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.9   | 9.5   | 9.9   | 10.6  | 11.6  | 11.7  | 12.3  | 12.7  | 11.4  | 10.3  | 8.8   | 7.6   |  |  |
|                                                                        | 5% DB       | MCDBR  | 12.9  | 14.5  | 14.9  | 14.2  | 14.9  | 15.0  | 16.2  | 16.2  | 15.6  | 14.5  | 13.2  | 11.7  |  |  |
|                                                                        |             | MCWBR  | 8.2   | 8.8   | 7.7   | 6.6   | 6.3   | 5.7   | 5.5   | 5.1   | 5.8   | 7.2   | 7.7   | 7.8   |  |  |
|                                                                        | 5% WB       | MCDBR  | 12.2  | 13.7  | 13.7  | 12.1  | 12.8  | 13.6  | 13.3  | 13.6  | 13.5  | 13.2  | 11.6  | 10.3  |  |  |
|                                                                        |             | MCWBR  | 8.1   | 8.5   | 7.3   | 6.2   | 5.9   | 5.7   | 5.5   | 5.3   | 5.6   | 6.8   | 7.1   | 7.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.345 | 0.378 | 0.423 | 0.481 | 0.456 | 0.461 | 0.468 | 0.481 | 0.447 | 0.405 | 0.371 | 0.347 |  |  |
|                                                                        | taud        |        | 2.281 | 2.201 | 2.121 | 1.973 | 2.118 | 2.146 | 2.137 | 2.096 | 2.153 | 2.251 | 2.299 | 2.286 |  |  |
|                                                                        | Ebn at Noon |        | 766   | 798   | 807   | 788   | 822   | 818   | 807   | 780   | 775   | 760   | 728   | 723   |  |  |
|                                                                        | Edh at Noon |        | 88    | 113   | 138   | 173   | 154   | 150   | 150   | 151   | 133   | 106   | 87    | 81    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.53  | 2.32  | 3.41  | 4.67  | 5.79  | 6.33  | 6.48  | 5.83  | 4.28  | 2.78  | 1.67  | 1.27  |  |  |
|                                                                        | RadStd      |        | 0.22  | 0.30  | 0.30  | 0.50  | 0.46  | 0.46  | 0.47  | 0.35  | 0.33  | 0.30  | 0.21  | 0.17  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (41 neighbors) |  | +0.44 | N/A     | N/A    | N/A     | +0.45 | N/A   | N/A         | -130    | +72     | N/A     |

Nomenclature: See separate page